Thin perfect matchings [HOG26]

Proof overview of the Lean 4/Mathlib formalisation

Library `ThinMatching/` (Theorem 35 of the main paper)

Paper. A. Haqi, S. Oveis Gharan. *On Thin Perfect Matchings up to Polylogarithmic Factors*, arXiv:2606.01330v2 (2026) [HOG26]. Statement, section, algorithm and equation numbers below refer to [HOG26]; its numbering is independent of the main paper. The main paper (J. Breitung, T. Gervens, M. Grohe, *Graph Similarity Meets Optimal Transport*, `Graph_Similarity_Meets_Optimal_Transport.pdf`) cites [HOG26, Theorem 1.4] as its **Theorem 35** (p. 32, l. 1021–1022, App. E.2.1) and uses it, through Theorem 38, for the multiplicative upper bound of Theorem 11 (cut distance). This document is for a reviewer who wants to check that Theorem 35 is proved in Lean, conditionally on the tree-cut sparsifier theorem of Räcke and Shah [RS14]. The line-by-line correspondence table is `ThinMatching/HOG26_correspondence.pdf`.

Name clashes. The Lean names follow [HOG26]: `ThinMatching.Lemma36` is [HOG26, Lemma 3.6] (not Lemma 36 of the main paper), `ThinMatching.theorem13_support` is [HOG26, Theorem 1.3] (not the main paper's Theorem 11, whose Lean name is `GsimOT.theorem13_of_rackeShah`), and `ThinMatching.theorem14_bip` is [HOG26, Theorem 1.4].

1 Statements

1.1 Theorem 35 of the main paper

Thinness (main paper, Eq. (23), p. 32, l. 1014–1016): a permutation matrix $P \in \mathrm{Perm}(n)$ is α-thin with respect to $Q \in \mathrm{DS}(n)$ if

$$\sum_{(i,j)\in(X\times\overline{Y})\cup(\overline{X}\times Y)} P_{ij} \;\le\; \alpha \sum_{(i,j)\in(X\times\overline{Y})\cup(\overline{X}\times Y)} Q_{ij} \qquad \text{for all } X, Y \subseteq [n].$$

Theorem 35 ([HOG26, Theorem 1.4]): there is an absolute constant $C_{\mathrm{thin}} > 0$ such that every $Q \in \mathrm{DS}(n)$ admits an α-thin permutation matrix with $\alpha \le C_{\mathrm{thin}} \ln(2n) \ln\ln(4n)$.

In Lean (main library, `GsimOT/Theorem13_CutUpperBound/Definitions.lean`) this is the proposition `GsimOT.ThinPermutationTheorem`, with thinness `GsimOT.IsThin`: for $\pi : V \simeq W$ and all $U_V \subseteq V$, $U_W \subseteq W$,

$$\#\{v : \neg(v \in U_V \leftrightarrow \pi(v) \in U_W)\} \le a \sum_{v,w} [\neg(v \in U_V \leftrightarrow w \in U_W)]\, Q_{vw},$$

which is Eq. (23) with $X = U_V$, $Y = U_W$: the pair (v, w) is counted iff $(v, w) \in (X \times \overline{Y}) \cup (\overline{X} \times Y)$. `ThinPermutationTheorem` asks for one $C_{\mathrm{thin}} > 0$ such that for all finite types V, W with $|V| = |W|$ and all Q with `IsDS Q` there is π with `IsThin` $(C_{\mathrm{thin}} \ln(2|V|) \ln\ln(4|V|))\, Q\, \pi$. This is exactly Theorem 35 (with V, W of size n instead of $[n]$; "$\alpha \le$" is equivalent to "$\alpha =$" because the right-hand side of Eq. (23) is nonnegative). For $n = 0$ the statement is trivial; for $n \ge 1$ the bound is positive, as required by "Let $\alpha > 0$". So `GsimOT.ThinPermutationTheorem` is Theorem 35.

1.2 Formalised statements of [HOG26]

Throughout, $x(\delta(S))$ = `cutWeight x S`, $|M \cap \delta(S)|$ = `crossCount M S`, and `IsThin` $\alpha\, x\, M$ means $|M \cap \delta(S)| \le \alpha\, x(\delta(S))$ for all $S \subseteq V$ (`Basic.lean`; [HOG26] quantifies over $\emptyset \ne S \subsetneq V$, which is equivalent since both sides vanish for $S \in \{\emptyset, V\}$). $x \in \mathrm{PM}(G)$ is `IsFracPM x`: a convex combination of perfect matchings. In the bipartite case $x : L \to R \to \mathbb{R}$ with `IsFracPMBip x` (nonnegative, all row and column sums 1, i.e. x is doubly stochastic), $\pi : L \simeq R$, and `IsThinBip`.

Theorem 1.4, bipartite `ThinMatching.theorem14_bip` (hRS : `RackeShahTheorem`): there is $C > 0$ such that for all finite L, R with $|L| = |R| = n$ and every x with `IsFracPMBip x` there is $\pi : L \simeq R$ (not necessarily inside $\mathrm{supp}(x)$) with $|M_\pi \cap \delta(S)| \le C \ln(2n) \ln\ln(4n)\, x(\delta(S))$ for all $S = S_L \cup S_R$. **This is the statement used for Theorem 35.**

Theorem 1.4, general `ThinMatching.theorem14` (hRS): there is $C > 0$ such that every $x \in \mathrm{PM}(G)$ on V has a $C \ln(2|V|) \ln\ln(4|V|)$-thin perfect matching M of the complete graph on V $(= C \ln(4n) \ln\ln(8n)$ for $|V| = 2n)$.

Theorem 1.3 `ThinMatching.theorem13_support` (h36 : `Lemma36`): there is $C > 0$ such that every $x \in \mathrm{PM}(G)$ has a perfect matching $M \subseteq \mathrm{supp}(x)$ (`InSupport`) that is $C|V| \ln(2|V|)$-thin.

Lemma 1.2 `ThinMatching.lemma12` (no hypotheses): for $n \geq 1$ the graph of Fig. 1 on $2n+2$ vertices has $x \in \mathrm{PM}(G)$ such that every perfect matching $M \subseteq \mathrm{supp}(x)$ has a cut $\emptyset \neq S \subsetneq V$ with $x(\delta(S)) > 0$ and $|M \cap \delta(S)| \geq \frac{n}{2}\, x(\delta(S))$; corollary `lemma12_not_thin`: no such M is α-thin for $\alpha < n/2$.

Lemma 1.2 and Theorem 1.3 are not needed for Theorem 35; they are formalised for completeness.

1.3 The two explicit hypotheses

- `RackeShahTheorem` (`TreeSparsifier.lean`) = [HOG26, Theorem 2.2] = [RS14]: there is $C > 0$ such that every weighted graph x on $n \geq 1$ vertices has a $C \ln(2n) \ln\ln(4n)$-tree-cut sparsifier (`IsTreeCutSparsifier`, Definition 2.1: a weighted tree T with Steiner nodes, $V \hookrightarrow \mathcal{V}$, and $x(\delta(S)) \leq \lambda_T(S) \leq q\, x(\delta(S))$ for all S). [HOG26] cites this result without proof; its proof (Räcke–Shah, building on Räcke's hierarchical decompositions) is far outside the scope of this formalisation. $\ln(2n) \ln\ln(4n)$ replaces $\log n \log\log n$ so that the bound is positive for $n \geq 1$.
- `Lemma36` (`SupportConstrained/Sampling.lean`) = [HOG26, Lemma 3.6], whose proof [HOG26] omits ("a simple Chernoff bound and a cut-counting argument"): there are $c, K > 0$ such that for every weighted graph $y \leq 1$ on W, every $N \geq \max(1, |W|)$ and every integer $\alpha \geq cN \ln(2N)$: if all nonempty proper cuts have $y(\delta(S)) \geq 1/(4N)$, then with probability $\geq 1 - 1/(4N)$ the sample Y (α independent copies of each edge e, each kept with probability y_e) satisfies $Y(\delta(S)) \leq K\alpha\, y(\delta(S))$ for all S. Only Theorem 1.3 uses it.

Both are Lean `def`s of type `Prop` that appear as arguments of the theorems, never as `axiom`s: the dependence is visible in every statement, a false hypothesis cannot make other parts of the library inconsistent, and `#print axioms` shows only `propext`, `Classical.choice`, `Quot.sound`. **Theorem 35 depends only on `RackeShahTheorem`, not on `Lemma36`.**

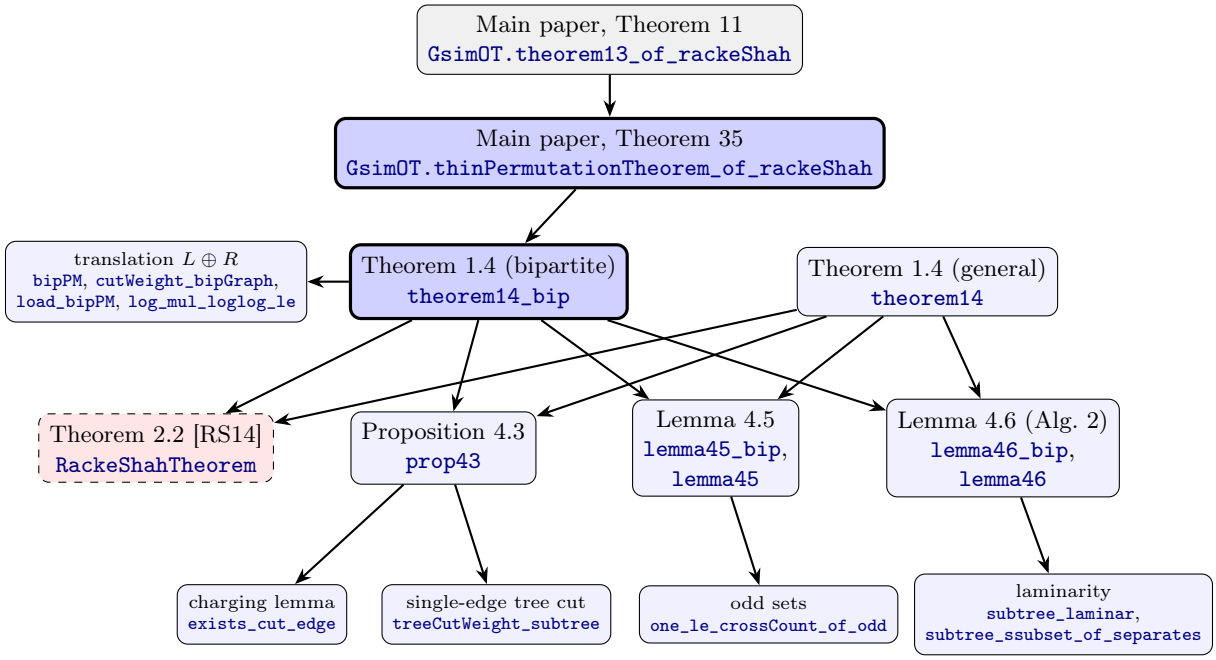
2 Where the proof lives

Paths are relative to `ThinMatching/` unless stated otherwise.

File	Content
`Basic.lean`	weighted graphs, cuts, perfect matchings (fixed-point-free involutions), thinness, $\mathrm{PM}(G)$; bipartite versions (Secs. 1, 2)
`TreeSparsifier.lean`	rooted trees, subtrees, tree cuts, λ_T, Definition 2.1, hypothesis `RackeShahTheorem` (Theorem 2.2)
`Load.lean`	loads and discrepancies (Definitions 4.1, 4.2, 4.4)
`LoadVsCut.lean`	Proposition 4.3 (`prop43`) and the charging lemma `exists_cut_edge`
`Discrepancy.lean`	Lemma 4.5 (`lemma45_bip`, `lemma45`)
`TreePairing.lean`	Algorithm 2, Lemma 4.6 (`lemma46_bip`, `lemma46`)
`Theorem14.lean`	Theorem 1.4 (`theorem14_bip`, `theorem14`); bipartite/general translation; constants
`LowerBound.lean`	Lemma 1.2, Fig. 1
`SupportConstrained/Prob.lean`	finite product spaces of Bernoulli coins (probability model of Algorithm 1)
`SupportConstrained/Sampling.lean`	the sampling step, hypothesis `Lemma36`
`SupportConstrained/Trimming.lean`	Definition 3.1, Lemma 3.2
`SupportConstrained/TutteWitness.lean`	Tutte witnesses (Definition 3.3, Theorem 2.4 via Mathlib's `SimpleGraph.tutte`), Lemma 3.4, Eq. (1)
`SupportConstrained/Lemma35.lean`	Lemma 3.5
`SupportConstrained.lean`	Theorem 1.3 (`theorem13_support`), Algorithm 1
`GsimOT/Theorem13_CutUpperBound/FromThinMatching.lean`	bridge (main library): Theorem 35 and Theorem 11 of the main paper assuming only [RS14]

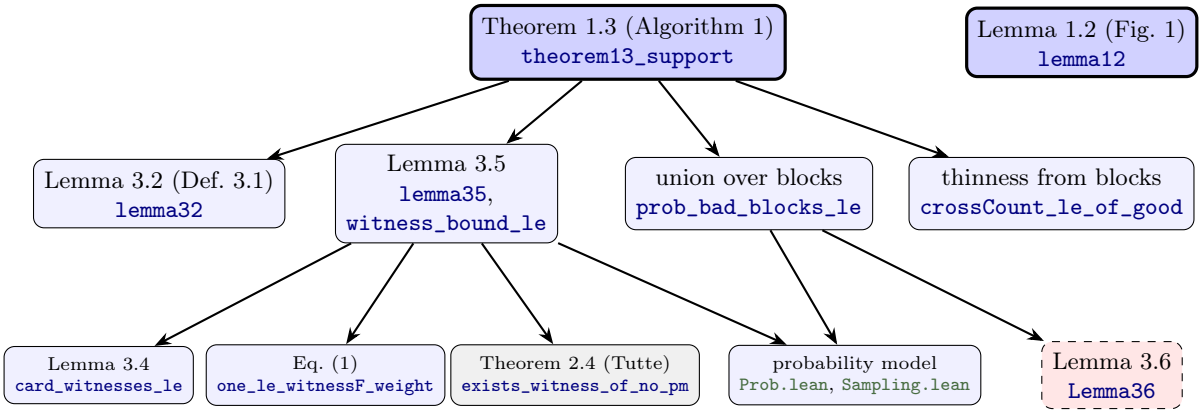
3 Proof layout

3.1 Theorem 1.4 and the bridge to the main paper



Theorem 11 of the main paper also needs the rest of `GsimOT/Theorem13_CutUpperBound` (Lemmas 36, 37, Theorems 38, 39, ... of the main paper, `GsimOT.theorem13_of_thin`); see the overview of that folder.

3.2 Theorem 1.3 and Lemma 1.2



Styles: thick = main results, dashed red = explicit hypotheses (cited results), grey = results outside this part (Mathlib's Tutte theorem; the main paper's Theorem 11).

4 The proof of Theorem 1.4 step by step

The proof of [HOG26, Section 4] has three ingredients and a short assembly.

Step 0: trees and tree-cut sparsifiers (Sec. 2, Definition 2.1, Theorem 2.2). A weighted tree with Steiner nodes is a `RootedTree N` (parent function, every vertex reaches the root), edge weights $y : N \to \mathbb{R}$ (edge $\{v, \mathrm{parent}(v)\}$ has weight y_v) and an embedding $e : V \hookrightarrow N$. `treeCutWeight` is $y(\delta_T(\mathcal{S}))$, `lambdaT` is $\lambda_T(S) = \min\{y(\delta_T(\mathcal{S})) : \mathcal{S} \cap V \in \{S, V \setminus S\}\}$, and `IsTreeCutSparsifier` q is Definition 2.1. Theorem 2.2 is the hypothesis `RackeShahTheorem`.

Step 1: loads and discrepancies (Definitions 4.1, 4.2, 4.4; `Load.lean`). The tree path between a and b uses the edge $f_v = \{v, \mathrm{parent}(v)\}$ iff exactly one of a, b lies in `subtree T v`. Hence $\mathrm{load}_M(f_v)$ (`load`,

loadBip) counts the matching edges with exactly one endpoint in the subtree of v, and the discrepancies are $\Delta_{\text{bip}} = \big||L \cap \cdot| - |R \cap \cdot|\big|$ (discBip) and $\Delta_{\text{non-bip}} = \mathbf{1}[|V \cap \cdot| \text{ odd}]$ (discNonBip) of $V \cap \text{subtree}(v)$.

Step 2: Proposition 4.3 (p. 7; prop43). If T is a q-tree-cut sparsifier of x and $\text{load}_M(f_v) \le \beta\, x(\delta(V \cap \text{subtree}(v)))$ for every tree edge, then M is βq-thin. Lean follows the paper: take an optimal tree cut $\mathcal{S}'$ for S (exists_eq_ciInf_of_finite); every crossing matching edge is charged to an edge $f_v \in \delta_T(\mathcal{S}')$ on its tree path (exists_cut_edge); since the single-edge cut subtree(v) has value y_v (treeCutWeight_subtree) and the sparsifier does not underestimate cuts, $\text{load}_M(f_v) \le \beta x(\delta(\cdot)) \le \beta \lambda_T(\cdot) \le \beta y_v$; summing, $|M \cap \delta(S)| \le \beta\, y(\delta_T(\mathcal{S}')) = \beta \lambda_T(S) \le \beta q\, x(\delta(S))$.

Step 3: Lemma 4.5 (p. 8; Discrepancy.lean). Bipartite (lemma45_bip): by the degree constraints, $\big||S_L| - |S_R|\big| = |x(S_L, R) - x(L, S_R)| \le x(\delta(S))$, as in the paper. General (lemma45): $x(\delta(S)) \ge 1$ for $|S|$ odd; Lean writes x as a convex combination of perfect matchings, each of which crosses an odd set (one_le_crossCount_of_odd, cutWeight_eq_sum_crossCount), instead of invoking the odd-set constraint (Theorem 2.5).

Step 4: Lemma 4.6, Algorithm 2 (pp. 9–10; TreePairing.lean). For points placed at tree vertices there is a perfect matching with $\text{load}_M(f) = \Delta(f)$ for every tree edge (lemma46_bip for $|L| = |R|$, lemma46 for $|V|$ even). Lean runs the bottom-up greedy one pair at a time: it matches a matchable pair inside a subtree of minimal size that contains one (the first pair the greedy forms) and recurses (strong induction on the number of points). The paper's key observation (inside the chosen subtree no proper subtree contains a matchable pair) becomes subtree_laminar, subtree_ssubset_of_separates: every edge separating the pair sees load and discrepancy both grow by one, all other edges see no change.

Step 5: assembly (Section 4; Theorem14.lean). theorem14: obtain the sparsifier T of x from RackeShahTheorem with $q = C \ln(2|V|) \ln\ln(4|V|)$, the matching M of lemma46 ($|V|$ is even because a perfect matching exists), check the hypothesis of prop43 with $\beta = 1$ by lemma45, and conclude that M is q-thin. theorem14_bip: the same argument on the weighted graph bipGraph x on $L \oplus R$, with the translations bipPM ($\pi \mapsto$ involution of $L \oplus R$), crossCount_bipPM, cutWeight_bipGraph, load_bipPM. Since $|L \oplus R| = 2n$ the sparsifier bound is $C \ln(4n) \ln\ln(8n) \le 6C \ln(2n) \ln\ln(4n)$ (log_mul_loglog_le), so theorem14_bip holds with constant $6C$, where C is the constant of RackeShahTheorem. The case $n = 0$ is trivial.

5 The proof of Theorem 1.3 (Section 3) in brief

theorem13_support follows Algorithm 1 with $n = |V|$, $\varepsilon = 1/(4n)$ and $\alpha = \lceil c' n \ln(2n) \rceil$, $c' = \max(c, 4)$: (1) trimming (Definition 3.1, lemma32): blocks that are ε-edge-connected, removed weight $\le (n-1)\varepsilon \le 1/4$; (2) sampling (sampleProb); (3) Lemma 3.5 (lemma35, via Tutte's theorem exists_witness_of_no_pm, at most $(n+1)^n$ witnesses card_witnesses_le, and $x(F) \ge 1$ one_le_witnessF_weight): no perfect matching with probability $\le 1/4$ (witness_bound_le); (4) Lemma36 on each block plus a union bound over $\le n$ blocks (prob_bad_blocks_le): some block bad with probability $\le 1/4$; (5) an outcome of positive probability avoiding both events (exists_not_of_prob_lt_one) gives $M \subseteq Y \subseteq \text{supp}(x)$, and crossCount_le_of_good (with sum_cut_eq_sum_blocks) gives thinness for every cut, also cuts through several blocks. Constant: $C = K(\max(c, 4) + 2)$.

6 The bridge to the main paper

GsimOT/Theorem13_CutUpperBound/FromThinMatching.lean (main library; it imports ThinMatching.Theorem14):

- GsimOT.thinPermutationTheorem_of_rackeShah (hRS : ThinMatching.RackeShahTheorem) : GsimOT.ThinPermutationTheorem. Proof: take C from theorem14_bip; for Q with IsDS Q the triple (nonneg, row_sum, col_sum) is IsFracPMBip Q; the resulting π is thin in the sense of GsimOT.IsThin after unfolding bipCrossCount, bipCutWeight (a simpa). Hence Theorem 35 holds with $C_{\text{thin}} = 6\, C_{\text{RS}}$, where C_{RS} is the constant of RackeShahTheorem.
- GsimOT.theorem13_of_rackeShah, GsimOT.theorem13_power_of_rackeShah: Theorem 11 of the main paper (both bounds) assuming only [RS14], by plugging the above into GsimOT.theorem13_of_thin and GsimOT.theorem13_power_of_thin.

7 Differences from [HOG26]

- *Representation.* Trees are rooted (parent function); this is no loss of generality. Loads and discrepancies are taken on the subtree side $V_s(f)$; the paper uses the root side $V_r(f)$, which has the same discrepancy since $|L| = |R|$ resp. $|V|$ is even. Perfect matchings are fixed-point-free involutions (general) or bijections $L \simeq R$ (bipartite). $\mathrm{PM}(G)$ is defined as the convex hull of perfect matchings (bipartite: degree constraints, equivalent by Birkhoff–von Neumann).
- *Constants.* $\log n \log\log n$ is written $\ln(2n)\ln\ln(4n)$ (positive for $n \geq 1$), as in the main paper's Theorem 35. The bipartite constant is $6C_{\mathrm{RS}}$ because the sparsifier lives on $2n$ vertices.
- *Lemma 4.5 (general)* uses only the easy direction of Edmonds' theorem (Theorem 2.5).
- *Lemma 4.6* is proved by matching one pair at a time in a minimal subtree, not by running the greedy vertex by vertex; the invariant is the paper's.
- *Theorem 1.3*: $\varepsilon = 1/(4|V|)$ (the paper's $1/(8n)$ for $|V| = 2n$); $x(F) \geq 1$ instead of $x(F) \geq 1/2$; trimming by block labellings (one new block per step); the paper argues thinness only per component, Lean also for cuts through several components (no loss). `Lemma36` is formulated with a global $N \geq |W|$ and failure probability $\leq 1/(4N)$, which is what the omitted proof gives and what the proof of Theorem 1.3 uses.
- *Not formalised*: Corollary 1.6 (metric distortion; needs [ACR23]) and Theorem 2.3 (algorithmic sparsifiers, not used).

No error in [HOG26] was found in the formalised parts.

8 How to check

From the repository root (with `elan` on the `PATH`):

- `lake build ThinMatching` builds the library (it imports only Mathlib); `lake build GsimOT.Theorem13_CutUpperBound.FromThinMatching` builds the bridge.
- `bash build.sh` builds everything, greps for `sorry`/`admit`/`axiom`, and runs `scripts/CheckAxioms.lean`, which contains `#print axioms` for `ThinMatching.theorem14_bip`, `ThinMatching.theorem14`, `ThinMatching.theorem13_support`, `ThinMatching.lemma12`, the lemmas of Sections 3–4 and `GsimOT.thinPermutationTheorem_of_rackeShah`, `GsimOT.theorem13_of_rackeShah`. Each must report only `propext`, `Classical.choice`, `Quot.sound`.
- To review the statements: read the definitions in `Basic.lean` and `TreeSparsifier.lean`, the hypothesis `Lemma36` in `SupportConstrained/Sampling.lean`, the main theorems in `Theorem14.lean`, `SupportConstrained.lean`, `LowerBound.lean`, and the bridge with `GsimOT.IsThin`, `GsimOT.ThinPermutationTheorem` in `GsimOT/Theorem13_CutUpperBound/Definitions.lean`.